add heart diagram question

Subject: Biology | Chapter: blood circulation | Difficulty: hard

Question 1: A patient presents with abnormally high blood pressure and elevated levels of angiotensin II. Which of the following scientific methodologies would be MOST effective in determining the underlying cause, focusing on the renin-angiotensin-aldosterone system (RAAS)?

Options:
A) Observational study of patient's diet and lifestyle.
B) Controlled experiment manipulating angiotensin II levels in a cell culture.
C) Comparative study of RAAS activity in different animal models with induced hypertension.
D) Randomized controlled trial comparing different antihypertensive medications.

Question 2: How does the structure of the capillary endothelium facilitate efficient exchange of gases and nutrients between blood and tissues, considering the principles of diffusion and surface area?

Options:
A) Thick, multi-layered epithelium reduces diffusion distance.
B) Presence of tight junctions prevents leakage and maintains blood pressure.
C) Large diameter capillaries increase flow rate, maximizing exchange.
D) Thin, single-layered epithelium with fenestrations increases surface area and reduces diffusion distance.

Question 3: A new species of deep-sea fish is discovered with hemoglobin exhibiting an exceptionally high oxygen affinity. What ecological adaptation is MOST likely responsible for this characteristic?

Options:
A) High metabolic rate requiring high oxygen delivery.
B) Low oxygen levels in its deep-sea environment.
C) Predatory lifestyle requiring bursts of high energy.
D) Symbiotic relationship with oxygen-producing bacteria.

Question 4: The lymphatic system plays a crucial role in maintaining fluid balance and immune function. Which specific cellular component of the lymphatic system is primarily responsible for antigen presentation and initiating adaptive immune responses?

Options:
A) Lymphocytes
B) Macrophages
C) Dendritic cells
D) Plasma cells

Question 5: A researcher hypothesizes that a novel compound inhibits the sodium-potassium pump in cardiac muscle cells. What physiological effect would this MOST likely have on cardiac output, considering the pump's role in maintaining the resting membrane potential and action potential propagation?

Options:
A) Increased heart rate and stroke volume.
B) Decreased heart rate and stroke volume.
C) Increased heart rate and decreased stroke volume.
D) No significant effect on cardiac output.

Generated on: 3/9/2025, 9:56:11 pm